

TED(15)-4074

REVISION-2015

MODEL QUESTION PAPER

FOURTH SEMESTER DIPLOMA EXAMINATION IN
CHEMICAL ENGINEERING
PETROLEUM AND ENERGY ENGINEERING

Maximum marks: 100

Time : 3hrs

PART-A

(Maximum marks: 10)

- I
- a) List any four petroleum products obtained from crude oil refining
 - b) Define hydrocracking
 - c) state the drift theory behind the origin of coal
 - d) Define proximate analysis
 - e) list any four thermal applications of solar energy

5x2=10

PART-B

(Maximum marks: 30)

- II
- Answer any five of the following question. Each question carries 6 marks

- a) Identify the parts of a fractionating column 6
- b) Illustrate the function of Redwood viscometer 6
- c) summarise the precautions taken for the storage of coal 6
- d) Illustrate froth floatation process for cleaning of coal 6
- e) summarise the proximate analysis method used for analysing coal 6
- f) Illustrate construction and working of electrical furnace 6
- g) Draw the figure and identify the parts of solar absorption refrigeration system 6

PART-C

(Maximum marks: 60)

(Answer one full question from each unit. Each question carries 15 marks)

Module-1

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|-----|----|--|---|
| III | a) | Explain the fixed bed catalytic cracking process with a flow diagram | 8 |
| | b) | Explain the sweetening process used for gasoline | 7 |

Or

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|----|----|---|---|
| IV | a) | Explain the sulphuric acid alkylation process with a flow diagram | 8 |
| | b) | Explain the occurrence of uranium and thorium | 7 |

Module-II

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|---|----|---|---|
| V | a) | Explain the manufacture of coke using Otto Hoffman byproduct oven | 8 |
| | b) | Illustrate the manufacture of carburated water gas | 7 |

Or

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|----|----|---|---|
| VI | a) | Explain the properties of coal used for metallurgical processes and gas producers | 7 |
| | b) | Illustrate the manufacture of producer gas | 8 |

Module-III

- | | | | |
|-----|----|---|---|
| VII | a) | Compare the functions of regenerators and recuperators used for heat saving | 8 |
| | b) | Explain the fluidised bed combustion system with a sketch | 7 |

Or

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|------|----|--|---|
| VIII | a) | Explain the constructional details and working of open hearth furnace with a neat sketch | 8 |
| | b) | Illustrate the working of chain grate stoker | 7 |

Module- IV

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|----|----|---|---|
| IX | a) | Explain the working cylindrical parabolic concentrating collectors with a neat sketch | 8 |
| | b) | Explain the forced circulation water heating system with a neat diagram | 7 |

Or

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|---|----|--|---|
| X | a) | Illustrate the working of silicon cell | 7 |
| | b) | Explain the working of tidal electric power generators | 8 |